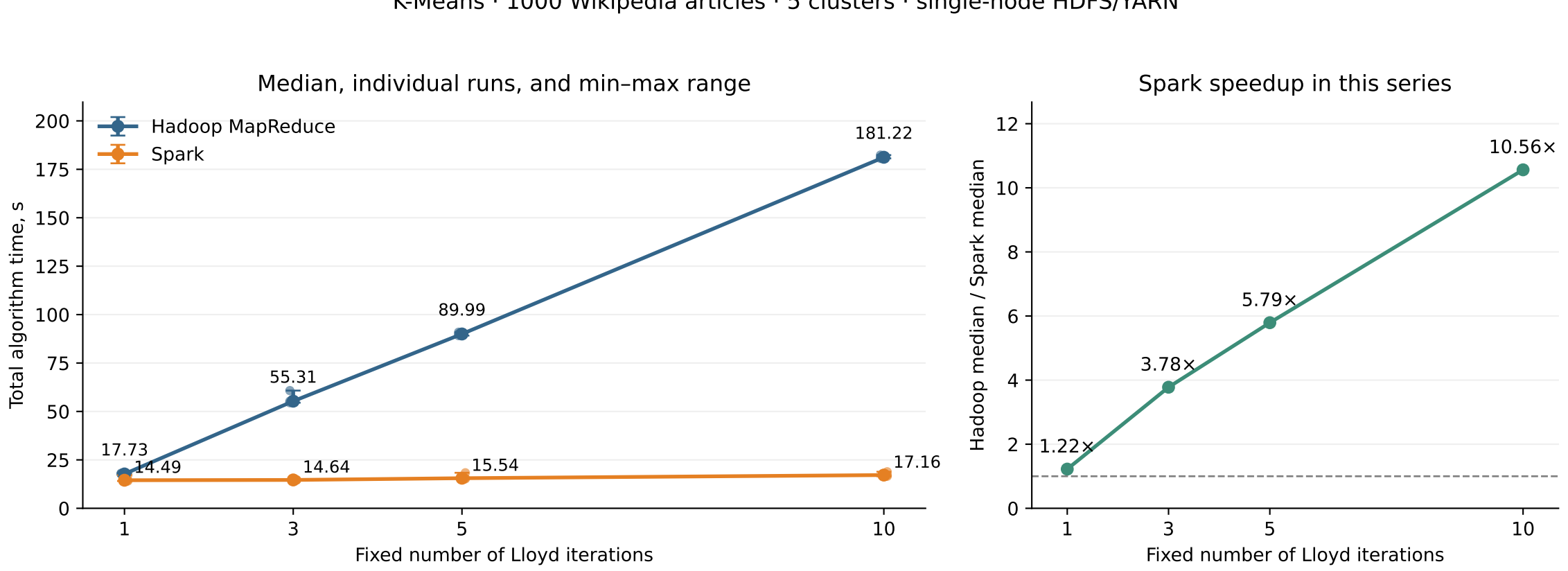
K-Means · 1000 Wikipedia articles · 5 clusters · single-node HDFS/YARN
Median, individual runs, and min–max range
Hadoop MapReduce
Spark
Total algorithm time, s
Fixed number of Lloyd iterations
181.22
89.99
55.31
17.73
14.49
14.64
15.54
17.16
Spark speedup in this series
Hadoop median / Spark median
Fixed number of Lloyd iterations
10.56×
5.79×
3.78×
1.22×
Complete algorithm time includes startup and initial reading; input preparation/upload excluded.